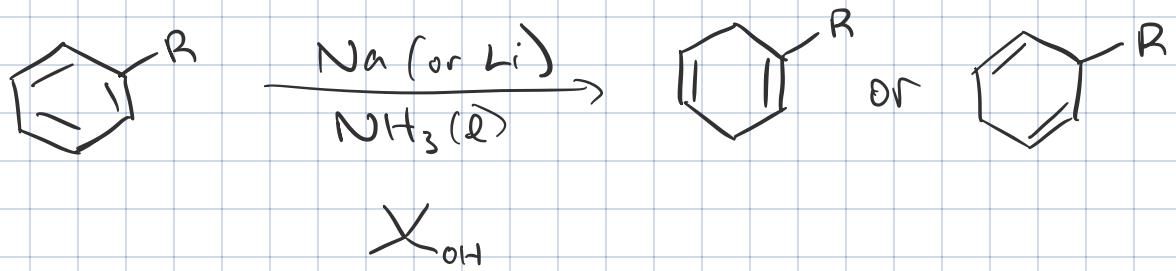
# Birch Reduction

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Reminder: Na & Li can dissolve in $NH_3(l)$ $-33°C$ to produce a blue solution via:

$$Na \rightarrow Na^+ + e^-$$

Since $NH_3$ has N (-3) it can't be reduced, and there are no free $H^+$ either. But polarity allows it to solvate charges

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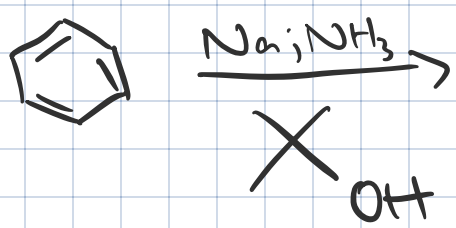
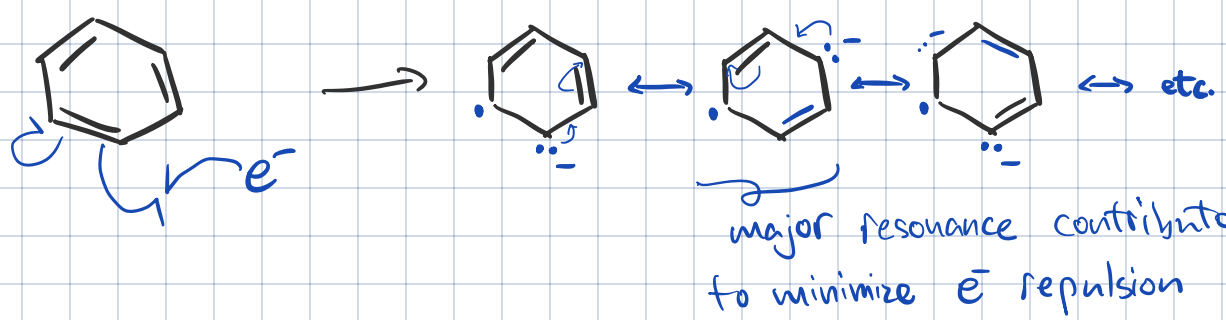
Birch Mechanism example




1. $Na \rightarrow Na^+ + e^-$

2. 

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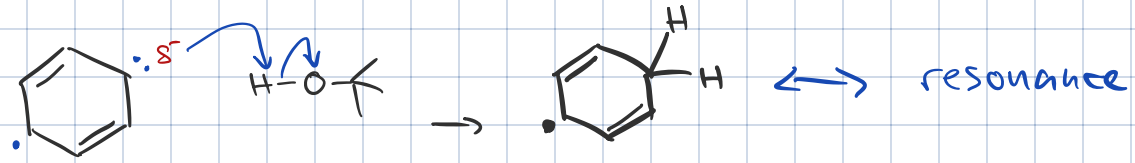


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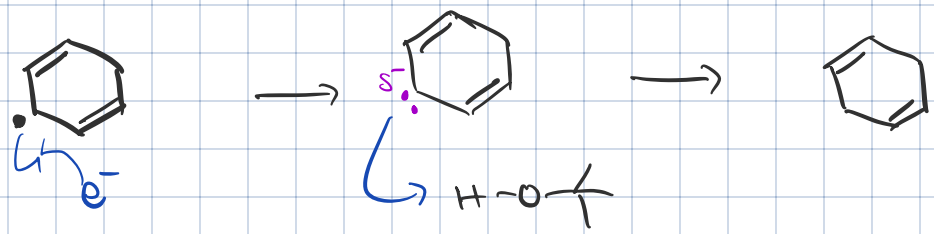


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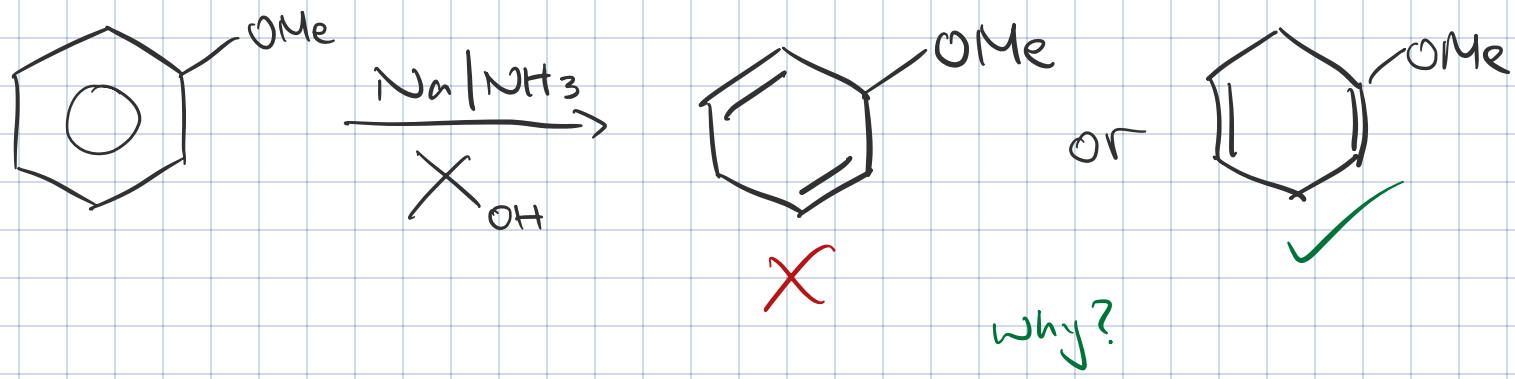


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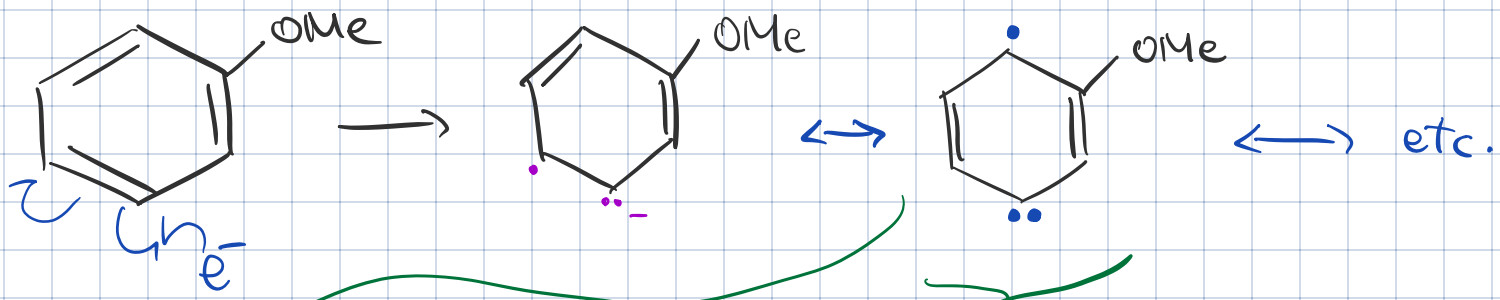
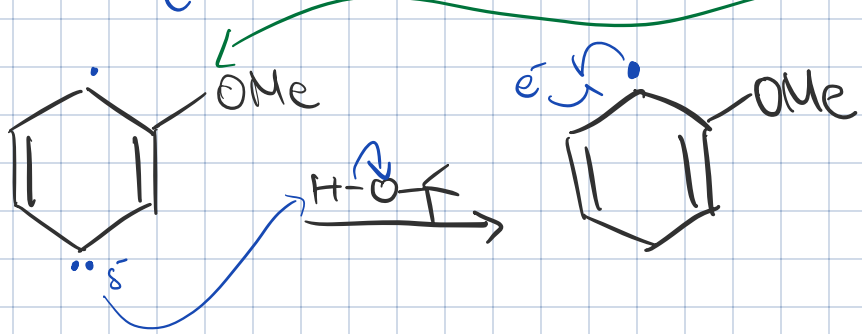
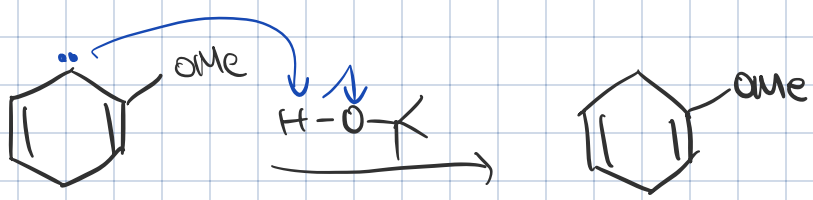
Mech w/ EDG




Major: → Recall that OMe donates e⁻ to o and p positions. We put the carbanion in the m position b/c it is the least e⁻ abundant. The radical goes on the other side to minimize e⁻ repulsion.




$sp^3$ C's not next to EDG b/c we couldn't put the carbanion intermediate (e⁻ repulsion)

Sidebar on Thermodynamics:

Why do $e^-$ rich aromatics accept even more $e^-$?

- The $e^-$ in solution are very high in energy, and even the LUMO $\pi^*$ of the aromats are much lower.

Why alcohols?

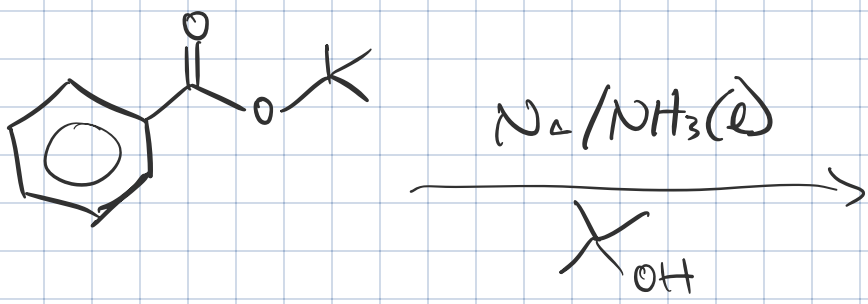
- Don't react w/ $e^-$ in solution (Acids stronger than $NH_4^+$ would produce $H^+ \rightarrow H\bullet \rightarrow \frac{1}{2}H_2$)
- Does react w/ carbanion intermediates

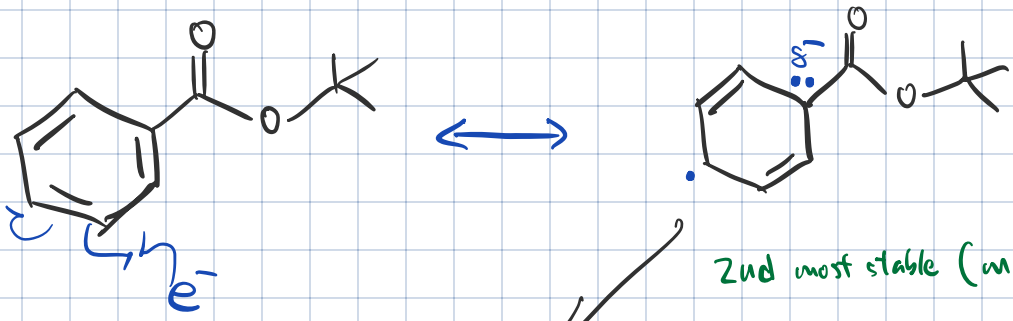
Why Tert-butanol?

- Tertbutoxide is a bulky NuPh → less side rxns
- Weaker acid than EtOH → controlled rxn

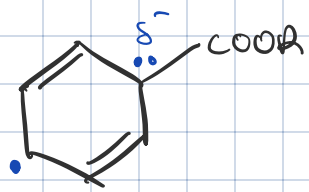
solvation of conjugate base

EWG Mech

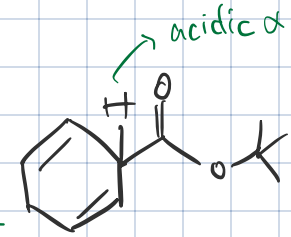


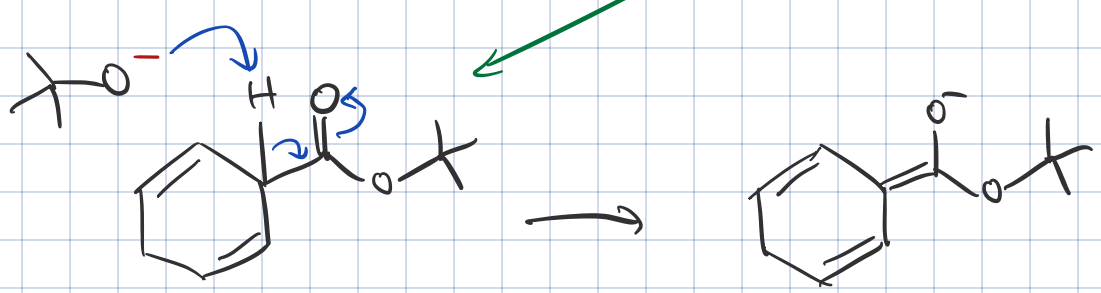




2nd most stable (major form is enolate)




Birch on 1,4 (see prev ex.)



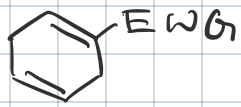


acid work-up to reform Birch product

## Regiochemistry of Birch Reduction

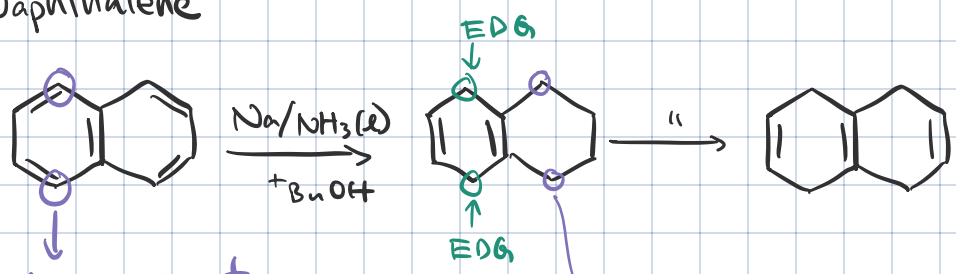
- Put carbanion at $e^-$-poor spots
- If there is an EWG, that C will end up w/ $sp^3$

- If no EWG but EDG, that C will be $sp^2$




## Cool Ex.

Naphthalene




only arrangement that preserves the other ring's aromaticity

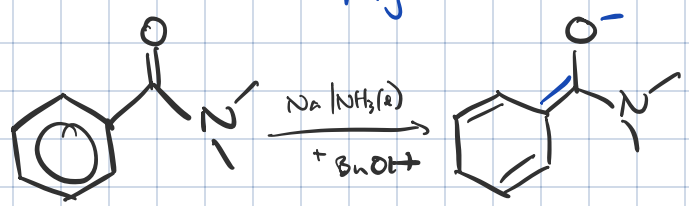
only way to put the anion & radical in para w/o putting one on the EDG carbon

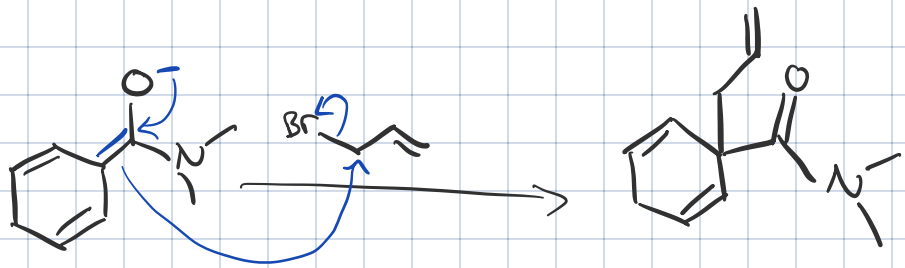
## Extension: Birch Alkylation

- Instead of doing the aforementioned $H^+$ workup for the EWG version, do a rxn w/ electrophile

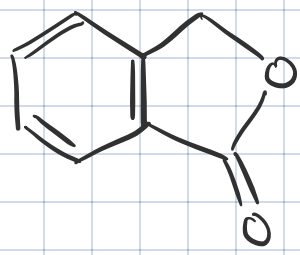
Ex.

Major Resonance after Step 1




Problem:



1. Na | $NH_3$ (l)
   + BuOH

2.

?

Sol